

Worked for 3m

The framework's core logic is:

Do not assume that more chips, more capex, or more tokens automatically mean AI is creating economic value. Trace the money from useful customer outcomes back to the infrastructure investment.

It is essentially a **public-investor underwriting bridge** between downstream AI adoption and upstream capital spending.

The simplest analogy

Think of AI infrastructure as a toll road:

- **AI capex** = money spent building the road
- **Tokens** = cars traveling on the road
- **Paid useful work** = paying cars that actually reach a valuable destination
- **AI revenue** = tolls collected
- **Contribution profit** = toll revenue minus power, cloud, model, support, and other direct operating costs
- **Capital charge** = depreciation, maintenance, and the return investors require for funding the road

A busy road is not necessarily a good investment. Cars may travel for free, tolls may be too low, or construction may have been far too expensive.

The framework asks:

Is the profit generated by paid AI activity sufficient to justify the capital investment?

The full pipeline

```
SEC financial statements
+
Manual AI-specific disclosures
↓
Low / Base / Upside estimates
↓
Paid Useful Work score
+
AI capex and contribution-profit estimates
↓
AI capital-stock model
↓
Economic coverage + new-capex IRR
```

1. Start with objective company financial data

The code automatically downloads SEC data for companies such as:

- Nvidia
- Microsoft
- Amazon
- Alphabet
- Meta
- Oracle
- Palantir
- ServiceNow
- Adobe
- Intuit

It collects company-wide:

- Revenue
- Gross profit
- Operating cash flow
- Capital expenditures
- Depreciation
- Accounts receivable
- Net property, plant, and equipment
- Research and development

It then calculates things such as:

Capex growth

Gross-profit growth

$$\frac{\text{Capex}}{\text{Operating cash flow}}$$

$$\frac{\Delta \text{Gross profit}}{\Delta \text{Capex}}$$

These are broad warning indicators.

For example, if a company's capex is growing 70% while its gross profit is growing 10%, that does not prove the AI investment is bad. However, it tells you the company has not yet demonstrated that profits are catching up.

This SEC portion is automatic and relatively reliable.

2. Add the AI-specific information manually

The problem is that companies generally do not disclose:

- Exactly how much capex is AI-related
- Exact AI revenue
- AI contribution profit
- AI gross margin
- AI customer renewal rates
- Production versus experimental usage
- Customer ROI

Therefore, the framework asks you to enter those disclosures or estimates in:

```
data/manual_ai_metrics.csv
```

For example, for Microsoft you might enter:

```
AI capex share: 65%  
AI revenue run rate: estimated range  
AI contribution margin: estimated range  
Production share: estimated range
```

For Anthropic, you might enter:

```
AI revenue run rate  
Paid enterprise customer growth  
Production usage share  
Customer retention  
Contribution-margin estimate
```

For Palantir, you might enter:

```
AI-related revenue growth proxy  
Net revenue retention  
Production deployments  
Customer-value growth  
Contribution-margin estimate
```

Every observation contains:

- Low estimate
- Base estimate
- High estimate
- Confidence score

- Source date
- Source name
- Whether it was reported, derived, or modeled

This prevents one uncertain assumption from appearing to be a hard fact.

3. Build the Paid Useful Work score

The framework does **not** claim that tokens equal useful work.

Instead, it constructs a 0–100 proxy from six components:

Component	Weight	What it is trying to determine
Paid demand	25%	Are customers actually paying more?
Production adoption	20%	Is AI in real workflows rather than pilots?
Task success	15%	Does the AI successfully complete the task?
Persistence	20%	Are customers renewing and expanding?
Customer value	10%	Is the economic benefit growing?
Contribution margin	10%	Does revenue exceed direct AI delivery cost?

Conceptually:

$\text{Paid Useful Work} = \text{Paid demand} \times \text{Production use} \times \text{Success} \times \text{Persistence} \times$

The actual code converts each component into a score and calculates a weighted average.

Why tokens receive less weight

Suppose token use grows 200%, but:

- Customers are receiving free credits
- Agents repeatedly call themselves without completing work
- Gross margins deteriorate
- Customers do not renew
- Most usage remains experimental

That is not strong economic demand.

Therefore, `paid_token_index` is allowed as a paid-demand proxy, but its score is capped. Token growth alone cannot make a company appear fully healthy.

Missing information is not treated as zero

Suppose you only have revenue growth and no information about:

- Renewals
- Production usage
- Task success
- Contribution margins

The framework does not assume all missing values are zero. It recalculates the weights using the available components and reports a separate **data-completeness score**.

However, it requires at least 35% of the total component weight to be observable. Otherwise, it returns:

`INSUFFICIENT_DATA`

That is deliberate. It is better than producing a confident but fabricated conclusion.

4. Estimate AI capex and AI contribution profit

For each public company, the code estimates quarterly AI capital spending using one of two methods:

$$\text{AI capex} = \text{Direct AI capex disclosure}$$

or:

$$\text{AI capex} = \text{Total company capex} \times \text{Estimated AI capex share}$$

It then estimates AI contribution profit using the best available input:

$$\text{Direct AI contribution profit}$$

or:

$$\text{AI revenue} \times \text{AI contribution margin}$$

or, for companies such as Meta where AI improves an existing business:

$$\text{Incremental company gross profit} \times \text{Estimated AI attribution share}$$

This last method is useful for businesses where there is no separately priced “AI product.”

For example, Meta’s AI may increase advertisement conversion and engagement. The value appears in advertising gross profit, not necessarily in a separate AI revenue line.

5. Turn capex into a capital stock

One dollar of capex is not consumed immediately. It creates an asset that should generate profit for several years.

The framework therefore puts every quarter’s AI capex into a separate **vintage**.

For example:

```
Q1 2026 capex vintage
Q2 2026 capex vintage
Q3 2026 capex vintage
Q4 2026 capex vintage
```

Each vintage is depreciated over an assumed economic life:

Scenario	Default AI asset life
Downside	3 years
Base	4 years
Upside	5 years

The downside assumes faster obsolescence. The upside assumes equipment remains economically useful longer.

For each period, the model calculates:

Economic depreciation

Required return on net capital

Maintenance capital charge

The default required return is your preferred:

15%

The total annual economic cost of the investment is:

$\text{Capital charge} = \text{Economic depreciation} + \text{Required return} + \text{Maintenance charge}$

This is more demanding than merely asking whether the equipment generates positive EBITDA.

6. Compare AI profit with the capital charge

The central ratio is:

$$\text{AI Economic Coverage} = \frac{\text{AI contribution profit}}{\text{Economic depreciation} + \text{Required return} + \text{Maintenance}}$$

The code interprets it approximately as:

Coverage	Meaning
At least 1.25×	SUPPORTED
Around 0.8×–1.25×	WATCH
Below 0.8× and not improving	NOT_SUPPORTED
Insufficient or low-confidence inputs	INSUFFICIENT_AI_DISCLOSURE

Hypothetical example

Suppose a hyperscaler has:

AI economic depreciation	\$10 billion
Required return charge	6 billion
Maintenance charge	2 billion

Total AI capital charge	\$18 billion

If estimated annual AI contribution profit is \$23 billion:

$$\frac{23}{18} = 1.28 \times$$

That would be classified as broadly supported.

If contribution profit is only \$10 billion:

$$\frac{10}{18} = 0.56 \times$$

The investment is not yet covering its full economic cost.

The company may still be making a rational long-term investment, but it has not yet demonstrated an adequate return.

7. Separately test the return on new capex

Economic coverage tells you whether the **existing installed capital base** is being supported.

It does not necessarily tell you whether the **next dollar of capex** will be attractive.

That is why the framework also builds a capex-cohort IRR.

For a new year's capex, it models:

- First-year utilization
- Second-year utilization
- Mature utilization
- Revenue per dollar of capital
- Contribution margin
- Annual compute-price decline
- Efficiency improvements
- Maintenance capex
- Taxes
- Useful life
- Salvage value

The simplified revenue calculation is:

$\text{AI infrastructure revenue} = \text{Capex} \times \text{Revenue per unit of capital} \times \text{Utilization}$

Then:

$\text{Cash contribution} = \text{Revenue} \times \text{Contribution margin} - \text{Maintenance} - \text{Taxes}$

The model calculates an IRR and compares it with:

15%

The output is:

- ABOVE_HURDLE
- MARGINAL
- BELOW_HURDLE

Why coverage and IRR can disagree

A company may have strong coverage today because existing GPUs were purchased cheaply and are rented at high prices.

But new capital might earn a poor return because:

- New systems cost more
- Rental prices are falling
- Utilization is declining
- Competition is increasing
- Hardware becomes obsolete faster

So you can see:

Current capital coverage: strong
New-capex IRR: below hurdle

That would mean the existing business is profitable, but management may be overinvesting at the margin.

8. Compare downstream strength with upstream economics

This is the main market-level framework.

There are two separate axes:

1. Is paid useful AI work growing?
2. Are capital owners earning adequate returns?

Downstream paid useful work	Capital-owner coverage	Interpretation
Strong	Strong	Healthy AI investment cycle
Strong	Weak	AI is useful, but infrastructure may be overbuilt or value may accrue mainly to applications
Weak	Strong	Infrastructure profits may depend on temporary scarcity or high pricing
Weak	Weak	Serious bubble or overbuild warning

The code classifies the overall ecosystem as:

- HEALTHY_CATCHUP
- MIXED
- OVERBUILD_WARNING
- INSUFFICIENT_DATA

An overbuild warning becomes more likely when:

Upstream capex growth $> 15\%$

while either:

AI economic coverage $< 0.8\times$

or:

Paid Useful Work score < 45

A healthy catch-up generally requires:

Capital-owner coverage $\geq 1.25\times$

and:

Downstream Paid Useful Work score ≥ 65

9. Avoid double-counting the same AI dollar

This is a very important design choice.

Suppose a business pays Palantir \$100.

Palantir pays a model provider \$10. The model provider pays Microsoft Azure \$5. Microsoft purchases Nvidia equipment.

You cannot add:

```
Palantir revenue
+ model-provider revenue
+ Azure revenue
+ Nvidia revenue
```

and call the result total AI economic demand. Much of it is the same money passing through the value chain.

The framework therefore keeps the layers separate:

End customer
↓
Application company
↓
Model provider
↓
Cloud provider
↓
Chip and infrastructure suppliers

It compares the strength of each layer rather than summing their revenue.

10. How the companies are used

The default company universe has different roles.

Upstream indicators

Nvidia
AMD
Broadcom
Micron

These tell you about infrastructure demand, pricing, and supply-chain economics.

Capital owners

Microsoft
Amazon
Alphabet
Meta
Oracle

These are the companies making enormous capital investments. The economic-coverage calculation is especially relevant to them.

Downstream demand indicators

OpenAI
Anthropic
Palantir
ServiceNow
Adobe
Intuit

These help answer whether the infrastructure is producing paid, recurring, economically useful work.

Meta belongs on both sides because it owns infrastructure and uses AI inside its advertising and application businesses.

11. How Jensen's supply forecast fits

The framework does not say Jensen Huang's supply-constrained forecast is useless.

It says his approach is most valuable for the **near term**:

$$\text{Nvidia revenue} = \min(\text{Available supply}, \text{Customer demand}) \times \text{Selling price}$$

If demand substantially exceeds supply, then knowing how many systems Nvidia can manufacture may be enough to forecast the next year's revenue.

But long-term demand becomes:

$$\text{Required compute} = \text{Paid useful work} \times \text{Compute required per unit of useful work}$$

Therefore:

- Jensen's supply model helps forecast FY2028 shipments.
- This framework tests whether enough downstream economics exist to support FY2029, FY2030, and beyond.

Both approaches can be correct at the same time.

What happens when you run the files

`run_pipeline.bat` launches `ai_monitor.py`.

The program:

1. Reads `config.yaml`.
2. Downloads SEC data.
3. Reads `manual_ai_metrics.csv`.
4. Creates downside, base, and upside cases.
5. Calculates Paid Useful Work scores.
6. Estimates AI capex and contribution profit.
7. Builds the AI capital stock.
8. Calculates economic coverage.

9. Calculates new-capex IRR.
10. Creates the company scorecard and ecosystem signal.
11. Writes the results into the `outputs` folder.

`dashboard.py` does not perform the main analysis. It simply reads those generated output files and displays them visually.

The most important limitation

The framework can automatically collect company-wide financials, but it cannot automatically know exactly how much of Microsoft's capex is AI-related or what Anthropic's true contribution margin is.

Those require investor estimates.

That is why the most important outputs are not only the score and coverage ratio, but also:

- Data completeness
- Input confidence
- Low/base/upside range
- Reported versus modeled classification

The framework is therefore not an automatic "AI bubble detector." It is a disciplined system for forcing you to answer:

What evidence supports this assumption, and how sensitive is the conclusion i

Its most useful result may sometimes be:

INSUFFICIENT_AI_DISCLOSURE

That tells you the market may be making a very confident valuation judgment using evidence that is not actually publicly observable.